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Operator Ordering Schemes and Covariant Path Integrals of Quantum and Stochastic Processes in Curved Space

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A method is given for the derivation of covariant path integral solutions of quantum and stochastic processes in curved space.
The correspondence between operator ordering schemes and ordinary functions in phase space is studied and applied to the explicit construction of a class of equivalent lattice representations of path integrals. It is shown that this class is uniquely related to a covariant functional integral. Some arguments for $R/12$ as the quantum mechanical curvature potential are given. Simple rules for nonlinear point transformations are stated. The connection with previous works is discussed.

1. Introduction

In the last years an ever increasing interest has been paid to the path integral method in various branches of physics, especially for two reasons. First, the possibility of performing transformations of the dynamical quantities by simply changing integration variables in the path integral has eminent virtues in all dynamical theories with constraints. Secondly, it is a promising starting point to go beyond perturbation theory.
In quantum field theory it is a very effective method for the quantization of various nonlinear theories (e.g. chiral theories, Yang-Mills gauge theories and gravity) and for the derivation of the Feynman rules in these theories. It allows for a very elegant derivation of Ward-Takahashi identities and of gauge independence in gauge theories. Furthermore, a path-integral collective coordinate method has been developed for the study of extended structures like solitons and instantons [1, 2].
In nonlinear non-equilibrium thermodynamics path integral representations of the distribution function of a complete set of macrovariables are now receiving an increased attention. Such a representation, being completely equivalent to the description in terms of a Fokker-Planck equation, can be used to generate the diagrammatic expansion for correlation and response functions for classical systems [3–5].
Often in the literature the path integral method is considered as of limited value only, with the understanding that all results derived in this approach are to be checked by a parallel analysis of the equations of motion. This point of view is due originally to Edwards and Gulyaev [6], who investigated the change from cartesian to polar coordinates in the path integral formalism. Their conclusion was that in the latter coordinates a noncovariant additional potential term is required. Recently, Mizrahi [7] and also Omote [8] have obtained a noncovariant potential term in the path integral expression by starting from a covariant Schrödinger equation. The necessity of the term in question was observed by Gervais and Jevicki [9] in the study of nonlinear point canonical transformations in path integrals. These authors altogether use the "midpoint rule" for the skeletonized path integral which is related to the Weyl ordering prescription of operator expressions. With respect to Mizrahi's paper it has been stressed by Dowker [10] that there is no preferred position for the "midpoint rule" and that an infinity of different functional integral representations exist, each of which is related to a particular operator ordering scheme. Similar conclusions have been drawn by Leschke and Schmutz [11].
In view of this one might conclude that the functional integral representation corresponding to one definite quantum theory is not unique and explicitly depends on the way of transforming an operator expression

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into a function. This, however, would imply a lot of severe implications in path integrals for theories with constraints and when nonlinear point canonical transformations have to be performed.

In this paper we show that such incisive restrictions of the path integral method are not really present. A covariant functional integral in configuration space is derived, independently of the applied operator ordering scheme. The formal functional integral is uniquely defined by the time slicing method and the requirement that in each time intervall the action principle holds. No reference to a particular lattice choice is required. The various representations with the additional potential terms mentioned above are obtained via an averaging procedure.

The functional integral is directly computed from the original definition of the propagator. Finite time propagation is considered as a folding together of many short time propagators and these are calculated by applying the $\sqrt{\varepsilon}$-scaling method of the momenta [12, 13]. This method amounts to the calculation of the semiclassical approximation and differs from the commonly used approximation $\exp(-i\mathbf{H}\varepsilon)\approx 1-i\mathbf{H}\varepsilon$ [7, 8, 10, 14]. The calculations are performed for a general operator ordering prescription parametrized by a continuous parameter γ. Antistandard ordering $(\mathbf{p},\mathbf{q})$, standard ordering $(\mathbf{q}\,\mathbf{p})$ and Weyl ordering are contained as special cases. By our calculations the results of de Witt [15] are confirmed and extended. As to quantum mechanics, some arguments for a definite curvature potential are given.

In Section 2 below we set up our notation and the covariant Schrödinger equation. In Section 3 the mapping of operators into functions is considered for a class of operator ordering schemes. Section 4 contains the calculation of the short time propagator. In Section 5 the action principle is exploited and a covariant functional integral is obtained. In Section 6 path integral solutions of the covariant Fokker-Planck equation are considered. Finally, in Section 7 some conclusions are drawn.

2. Covariant Quantum Dynamics

Let us review quantum mechanics for a particle moving in curved n-dimensional space with a natural metric g_{ij} [16]. The matrix g_{ij} is symmetric and possesses an inverse g^{ij} satisfying

$$g^{ik}g_{kj}=\delta^i_j. \tag{2.1}$$

The eigenstates $|q\rangle$ and $|p\rangle$ of the position and momentum operators $\mathbf{q}$ and $\mathbf{p}$ are normalized according to

$$\langle q|q'\rangle = g^{-1/2}(q)\,\delta(q-q');$$
$$\langle p|p'\rangle = \delta(p-p'), \tag{2.2}$$

and the coordinate representation of the momentum eigenstate is given by

$$\langle q|p\rangle = (2\pi)^{-n/2}g^{-1/4}(q)\exp(iq\cdot p), \tag{2.3}$$

where $q\cdot p=q_ip^i$. There follows from (2.2) and (2.3) the coordinate representation of the momentum operator

$$\langle q|\mathbf{p}_i|p\rangle = -ig^{-1/4}(q)\frac{\partial}{\partial q^i}g^{1/4}(q)\langle q|p\rangle. \tag{2.4}$$

The wave equation has the form

$$\mathbf{H}(\mathbf{p},q)\langle q|\Psi(t)\rangle = i\frac{\partial}{\partial t}\langle q|\Psi(t)\rangle, \tag{2.5}$$

and the wave function $\Psi(q,t)\equiv\langle q|\Psi(t)\rangle$ is a scalar of weight zero.

It is well known that the ordering of position and momentum operators in the quantum Hamiltonian $\mathbf{H}(\mathbf{p},\mathbf{q})$ can be uniquely fixed by requiring that $\mathbf{H}$ be invariant under the point canonical transformation [16, 8]

$$\mathbf{q}'^i=\mathbf{q}'^i(\mathbf{q});$$
$$\mathbf{p}'_i=\tfrac{1}{2}\left(\frac{\partial\mathbf{q}^j}{\partial q'^i}\mathbf{p}_j+\mathbf{p}_j\frac{\partial\mathbf{q}^j}{\partial q'^i}\right). \tag{2.6}$$

There follows the covariant quantum Hamiltonian

$$\mathbf{H}(\mathbf{p},\mathbf{q})=\tfrac{1}{2}\mathbf{g}^{-1/4}(\mathbf{p}_i-\mathbf{A}_i)\mathbf{g}^{ij}\mathbf{g}^{1/2}(\mathbf{p}_j-\mathbf{A}_j)\mathbf{g}^{-1/4}$$
$$+\mathbf{V}+\lambda\mathbf{R}, \tag{2.7}$$

where g^{ij}, $g\equiv|g_{ij}|$, A_i, V and R are functions of the coordinates q. A_i and V may be the vector and scalar potentials, respectively and R is the curvature scalar

$$R=g^{ij}R_{ij}$$
$$R_{ij}=\Gamma^l_{ij,\,l}-\Gamma^l_{il,\,j}+\Gamma^l_{ij}\Gamma^m_{ml}-\Gamma^l_{im}\Gamma^m_{lj}, \tag{2.8}$$

defined in terms of the Christoffel symbol

$$\Gamma^i_{jk}=g^{il}[l,jk];$$
$$[l,jk]=\tfrac{1}{2}(g_{lj,\,k}+g_{lk,\,j}-g_{jk,\,l}). \tag{2.9}$$

Subscripts following a comma denote partial derivatives with respect to the component indicated.

The purely quantum mechanical potential λR will be discussed in Section 5.

3. Generalized Weyl Correspondence

In this section we consider the correspondence between quantum mechanical operator $\mathbf{A}(\mathbf{p},\mathbf{q})$ and ordinary functions in phase space $a(p,q)$ (cf. [17]). Here we generalize a treatment of Mizrahi [7] to a class of operator ordering schemes. By expanding the operator $\mathbf{A}(\mathbf{p},\mathbf{q})$ in position and momentum eigenstates

$$\mathbf{A}(\mathbf{p},\mathbf{q})=\int dp'\,dp''\,dq'\,g^{1/2}(q')\,dq''\,g^{1/2}(q'')\cdot|p'\rangle\langle p'|q'\rangle\langle q'|\mathbf{A}(\mathbf{p},\mathbf{q})|q''\rangle\langle q''|p''\rangle\langle p''|, \tag{3.1}$$

making the change of variables of Jacobian unity

$$p'=p-\alpha u,\quad p''=p+\beta u$$
$$q'=q-\beta v,\quad q''=q+\alpha v \tag{3.2}$$

$$\alpha=\tfrac{1}{2}(1-\gamma),\quad \beta=\tfrac{1}{2}(1+\gamma),\quad -1\leqq\gamma\leqq 1 \tag{3.3}$$

and using (2.3) we get

$$\mathbf{A}(\mathbf{p},\mathbf{q})=(2\pi)^{-n}\int dp\,dq\,g^{1/2}(q)\cdot a_\gamma(p,q)\,\Delta_\gamma(p,q), \tag{3.4}$$

where

$$a_\gamma(p,q)=\int dv\,g^{1/4}(q-\beta v)\,g^{1/4}(q+\alpha v)\,e^{ip\cdot v}\cdot\langle q-\beta v|\mathbf{A}(\mathbf{p},\mathbf{q})|q+\alpha v\rangle \tag{3.5}$$

$$\Delta_\gamma(p,q)=g^{-1/2}(q)\int du\,e^{iq\cdot u}\cdot|p-\alpha u\rangle\langle p+\beta u|. \tag{3.6}$$

$a_\gamma(p,q)$ is called the generalized Weyl transform of the operator $\mathbf{A}(\mathbf{p},\mathbf{q})$. The correspondence is one-to-one. (We shall denote it by $\leftrightarrow$.) By calculation one finds, e.g.,

$$\mathbf{p}^m\mathbf{f}(\mathbf{q})\,\mathbf{p}^n\leftrightarrow\left(p-i\alpha\frac{\partial}{\partial q}\right)^m\left(p+i\beta\frac{\partial}{\partial q}\right)^n f(q). \tag{3.7}$$

It is convenient to define a class of ordering schemes, denoted by $\mathscr{Z}_\gamma\{\mathbf{A}(\mathbf{p},\mathbf{q})\}$, such that the following simple correspondence holds

$$\mathscr{Z}_\gamma\{\mathbf{A}(\mathbf{p},\mathbf{q})\}\leftrightarrow A(p,q). \tag{3.8}$$

By definition, $\mathscr{Z}_\gamma$ is an operator which orders an operator function $\mathbf{A}(\mathbf{p},\mathbf{q})$ regardless of the commutation relation such that the correspondence (3.8) holds. There follows with (3.7)

$$\mathscr{Z}_\gamma\{\mathbf{p}^m\mathbf{f}(\mathbf{q})\}=\sum_{l=0}^{m}\binom{m}{l}(\beta\mathbf{p})^{m-l}\mathbf{f}(\mathbf{q})(\alpha\mathbf{p})^l, \tag{3.9}$$

which for the special cases $\gamma=1,0,-1$ represents the well known antistandard $(\mathbf{p}\,\mathbf{q})$, Weyl and standard $(\mathbf{q}\,\mathbf{p})$ ordering schemes.

In terms of (3.9) the Hamiltonian (2.7) assumes the form

$$\mathbf{H}(\mathbf{p},\mathbf{q})=\mathscr{Z}_\gamma\{\mathbf{d}(\mathbf{p},\mathbf{q})+\mathbf{b}_\gamma(\mathbf{p},\mathbf{q})+\mathbf{c}_\gamma(\mathbf{q})\}, \tag{3.10}$$

where

$$d(p,q)=\tfrac{1}{2}p_ip_jg^{ij}$$
$$b_\gamma(p,q)=p_lG^l_\gamma;\qquad G^l_\gamma=i\frac{\gamma}{2}g^{lm}_{,m}-A^l$$
$$c_\gamma(q)=V+(\lambda-\tfrac{1}{8})R+\tfrac{1}{8}g^{ij}\Gamma^m_{li}\Gamma^l_{jm}-\frac{\gamma^2}{8}g^{ij}_{,ij}+\tfrac{1}{2}A_iA_jg^{ij}-i\frac{\gamma}{2}A^j_{,j}. \tag{3.11}$$

For later convenience we give a relation which connects two differently ordered operator functions $\mathbf{A}_\gamma$ and $\mathbf{A}_{\gamma'}$

$$\mathscr{Z}_{\gamma'}\{\mathbf{A}_{\gamma'}(\mathbf{p},\mathbf{q})\}=\mathscr{Z}_\gamma\{\mathbf{A}_\gamma(\mathbf{p},\mathbf{q})\}. \tag{3.12}$$

There follows from (3.9) that the function $A_{\gamma'}(p,q)$ is related to $A_\gamma(p,q)$ by

$$A_{\gamma'}(p,q)=A_\gamma\left(p+\frac{i}{2}(\gamma'-\gamma)\frac{\partial}{\partial q},q\right), \tag{3.13}$$

with the understanding, that the differentiation $\partial/\partial q$ is to be performed on all q-dependent terms of A_γ. Finally, let us consider matrix elements in coordinate space. There follows from (3.6) and (2.3)

$$\langle q''|\Delta_\gamma(p,q)|q'\rangle=g^{-1/4}(q'')\,g^{-1/4}(q')\,g^{-1/2}(q)\,\delta(q-q_\gamma)\,e^{ip\cdot(q''-q')}, \tag{3.14}$$

where

$$q_\gamma=\tfrac{1}{2}(q'+q'')-\frac{\gamma}{2}(q''-q'). \tag{3.15}$$

Using (3.14) and (3.8) one gets from (3.4)

$$\langle q''|\mathscr{Z}_\gamma\{\mathbf{A}(\mathbf{p},\mathbf{q})\}|q'\rangle=(2\pi)^{-n}g^{-1/4}(q'')\,g^{-1/4}(q')\cdot\int dp\,e^{ip\cdot(q''-q')}A(p,q_\gamma). \tag{3.16}$$

In the next section (3.16) will be applied to the calculation of the short time propagator

4. Lattice Representations of Path Integrals

Let us now calculate the propagator of the Schrödinger equation

$$K(q_b,t_b;q_a,t_a)=\langle q_b|\mathbf{U}(\mathbf{p},\mathbf{q},t_b-t_a)|q_a\rangle, \tag{4.1}$$

where $\mathbf{U}$ is the time evolution operator

$$\mathbf{U}(\mathbf{p},\mathbf{q},t)=\exp[-i\mathbf{H}(\mathbf{p},\mathbf{q})t]. \tag{4.2}$$

There follows from (3.16) that if the ordering of $\mathbf{U}$ is performed according to

$$\mathbf{U}(\mathbf{p},\mathbf{q},t)=\mathscr{Z}_\gamma\{\mathbf{X}_\gamma(\mathbf{p},\mathbf{q},t)\}, \tag{4.3}$$

the general solution of (4.1) is reduced to quadratures

$$K(q_b,t_b;q_a,t_a)=(2\pi)^{-n}g^{-1/4}(q_b)\,g^{-1/4}(q_a)$$
$$\cdot\int dp\,e^{ip\cdot(q_b-q_a)}X_\gamma(p,q_\gamma,t_b-t_a). \tag{4.4}$$

However, the evaluation of $X_\gamma(p,q,t)$ for arbitrary t is an arduous task. Therefore, it is convenient to proceed in three steps. First, (4.3) is calculated in an appropriate approximation and applied to short time propagation. Secondly, the passage to finite time propagation is performed by a folding together of many short time propagators $K_{s\gamma}$

$$K(q_b,t_b;q_a,t_a)$$
$$=\lim_{N\to\infty}\int\prod_{j=1}^{N-1}\{dq_j\,g^{1/2}(q_j)K_{s\gamma}(q_{j+1},t_{j+1};q_j,t_j)\}$$
$$\cdot K_{s\gamma}(q_1,t_1;q_a,t_a), \tag{4.5}$$

where $t_j=t_a+\varepsilon j$, $\varepsilon=(t_b-t_a)/N$, $q_N=q_b$. When an explicit form of $K_{s\gamma}$ is inserted, (4.5) may be called the lattice representation of a path integral. In the last step a covariant path integral functional is traced.

Let us now consider the first step. It is well known that due to the term $\mathbf{d}(\mathbf{p},\mathbf{q})\varepsilon$, occuring in

$$\mathscr{Z}_\gamma\{\mathbf{X}_\gamma(\mathbf{p},\mathbf{q},\varepsilon)\}$$
$$=\exp[-i\varepsilon\mathscr{Z}_\gamma\{\mathbf{d}(\mathbf{p},\mathbf{q})+\mathbf{b}_\gamma(\mathbf{p},\mathbf{q})+\mathbf{c}_\gamma(\mathbf{q})\}], \tag{4.6}$$

the integration over p in (4.4) generates a $\varepsilon^{-n/2}$-singularity of the propagator. Following Graham [12, 13], we substitute $p=\tilde{p}/\sqrt{\varepsilon}$ in (4.4) in order to have a regular $\tilde{p}$-integral. There follows

$$K_{s\gamma}(q+\Delta,t+\varepsilon;q,t)$$
$$=(2\pi\sqrt{\varepsilon})^{-n}g^{-1/4}(q+\Delta)\,g^{-1/4}(q)$$
$$\cdot\int d\tilde{p}\,e^{i\tilde{p}\cdot\Delta/\sqrt{\varepsilon}}Y_\gamma(\tilde{p},q_\gamma,\varepsilon), \tag{4.7}$$

where

$$q_\gamma=q+\frac{(1-\gamma)}{2}\Delta \tag{4.8}$$

and $Y_\gamma(\tilde{p},q,\varepsilon)$ is obtained from (3.8) and the operator relation

$$\mathscr{Z}_\gamma\{\mathbf{Y}_\gamma(\tilde{\mathbf{p}},\mathbf{q},\varepsilon)\}$$
$$=\exp[-i\mathscr{Z}_\gamma\{\mathbf{d}(\tilde{\mathbf{p}},\mathbf{q})+\sqrt{\varepsilon}\,\mathbf{b}_\gamma(\tilde{\mathbf{p}},\mathbf{q})+\varepsilon\mathbf{c}_\gamma(\mathbf{q})\}], \tag{4.9}$$

where the new operator $\tilde{\mathbf{p}}=\sqrt{\varepsilon}\,\mathbf{p}$ obeys the commutation relation

$$(\tilde{\mathbf{p}}_i,\mathbf{f}(\mathbf{q}))=-i\sqrt{\varepsilon}\,\mathbf{f}_{,i}(\mathbf{q}). \tag{4.10}$$

We now calculate $Y_\gamma(\tilde{p},q,\varepsilon)$ as an expansion in $\sqrt{\varepsilon}$ up to terms of order ε

$$Y_\gamma(\tilde{p},q,\varepsilon)=\exp[-id(\tilde{p},q)]$$
$$\cdot\{1+\sqrt{\varepsilon}\,B_\gamma(\tilde{p},q)+\varepsilon C_\gamma(\tilde{p},q)+o(\varepsilon^{3/2})\}. \tag{4.11}$$

This is achieved from (4.9) in two steps. First, the case of antistandard ordering ($\gamma=1$) is calculated via recursion relations, paralleling a treatment of Graham [13]. We obtain

$$B_1(\tilde{p},q)=-ib_1-\frac{i}{2}\tilde{p}_l d_{,m}g^{lm}$$
$$C_1(\tilde{p},q)=\tfrac{1}{2}B_1^2-ic_1+\tfrac{1}{4}d_{,ij}g^{ij}-\frac{i}{2}d_{,j}G_1^j$$
$$-\frac{i}{2}b_{1,l}\tilde{p}_m g^{lm}-\frac{i}{6}(d_{,i}d_{,j}g^{ij}$$
$$+d_{,il}g^{ij}g^{lm}\tilde{p}_j\tilde{p}_m+d_{,i}g^{ij}_{,l}g^{lm}\tilde{p}_j\tilde{p}_m). \tag{4.12}$$

From this we proceed to our general class of ordering prescription by use of the relation

$$Y_\gamma(\tilde{p},q,\varepsilon)=Y_1\left(\tilde{p}+\frac{i}{2}\sqrt{\varepsilon}(\gamma-1)\frac{\partial}{\partial q},q,\varepsilon\right), \tag{4.13}$$

which follows from (3.13) and (4.10). We finally get

$$B_\gamma(\tilde{p},q)=-ib_\gamma-\frac{i}{2}\gamma\tilde{p}_l d_{,m}g^{lm},$$
$$C_\gamma(\tilde{p},q)=\tfrac{1}{2}B_\gamma^2-ic_\gamma-\frac{i}{2}\gamma d_{,j}G_\gamma^j$$
$$-\frac{i}{2}\gamma b_{\gamma,l}\tilde{p}_m g^{lm}+\tfrac{1}{24}g^{ij}[3(1+\gamma^2)d_{,ij}$$
$$-i(1+3\gamma^2)(d_{,i}d_{,j}+d_{,il}\tilde{p}_j\tilde{p}_m g^{lm})]$$
$$+\tfrac{1}{24}g^{ij}_{,l}[2i(1-3\gamma^2)d_{,i}\tilde{p}_j\tilde{p}_m g^{lm}$$
$$-3(1-\gamma^2)g^{lm}_{,j}\tilde{p}_i\tilde{p}_m], \tag{4.14}$$

where $d(\tilde{p},q)$, $b_\gamma(\tilde{p},q)$ and $c_\gamma(q)$ are defined in (3.11). Inserting (4.11) in (4.7) and shifting the integration variable according to $\tilde{p}_i=r_i+\bar{g}_{ij}\Delta^j/\sqrt{\varepsilon}$ in order to complete the square in the exponent we are left with a number of Gaussian integrals to be carried out

$$K_{s\gamma}(q+\Delta,t+\varepsilon;q,t)$$
$$=(2\pi\sqrt{\varepsilon})^{-n}\exp\left(\frac{i}{2\varepsilon}\bar{g}_{ij}\Delta^i\Delta^j\right)g^{-1/4}(q+\Delta)\,g^{-1/4}(q)$$
$$\cdot\int dr\exp\left(-\frac{i}{2}r_i r_j\bar{g}^{ij}\right)$$
$$\cdot[1+\sqrt{\varepsilon}\,\bar{B}_\gamma(r_i+\bar{g}_{ij}\Delta^j/\sqrt{\varepsilon})+\varepsilon\bar{C}_\gamma(r_i+\bar{g}_{ij}\Delta^j/\sqrt{\varepsilon})], \tag{4.15}$$

where $\bar{g}_{ij}=g_{ij}(q_\gamma)$ and q_γ is defined in (4.8). Next, the integrals in (4.15) are performed keeping all powers of $\Delta/\sqrt{\varepsilon}$ occuring in B_γ and C_γ and the terms $g^{-1/4}(q+\Delta)$ and $g^{-1/4}(q)$ are expanded about q_γ up to terms quadratic in Δ. We obtain after combining similar terms in a rather lengthy calculation the following expression of the short time propagator

$$
\begin{aligned}
&K_{s\gamma}(q+\Delta,t+\varepsilon;q,t)\\
&=(2\pi i\varepsilon)^{-n/2}\exp\left(\frac{i}{2\varepsilon}\bar{g}_{ij}\Delta^i\Delta^j\right)\\
&\cdot\left\{1+\tfrac{1}{12}R_{ij}\Delta^i\Delta^j+\frac{i}{4}\gamma\, g_{ij,l}\Delta^i\Delta^j\Delta^l/\varepsilon\right.\\
&+\frac{i}{48}((1+3\gamma^2)\,g_{ij,kl}-2g^{mn}\,[m,ij][n,kl])\\
&\cdot\Delta^i\Delta^j\Delta^k\Delta^l/\varepsilon+i\varepsilon(\tfrac{1}{12}-\lambda)R-i\varepsilon V\\
&+iA_i\Delta^i+\frac{i}{2}\gamma\,A_{i,j}\Delta^i\Delta^j\\
&\left.-\tfrac{1}{2}\left(\frac{\gamma}{4}g_{ij,l}\Delta^i\Delta^j\Delta^l/\varepsilon+A_i\Delta^i\right)^2\right\}_{q=q_\gamma}.
\end{aligned}
\tag{4.16}
$$

For later convenience we may write (4.16) in the following form

$$
\begin{aligned}
&K_{s\gamma}(q+\Delta,t+\varepsilon;q,t)\\
&\doteq(2\pi i\varepsilon)^{-n/2}(1+\tfrac{1}{12}\bar{R}_{ij}\Delta^i\Delta^j)\\
&\cdot\exp(iS_\gamma(q+\Delta,t+\varepsilon;q,t),
\end{aligned}
\tag{4.17}
$$

where

$$
\begin{aligned}
&S_\gamma(q+\Delta,t+\varepsilon;q,t)\\
&=\left\{\frac{1}{2\varepsilon}\left[g_{ij}\Delta^i\Delta^j+\frac{\gamma}{2}g_{ij,l}\Delta^i\Delta^j\Delta^l\right.\right.\\
&\left.+\tfrac{1}{24}((1+3\gamma^2)\,g_{ij,kl}-2g^{mn}\,[m,ij][n,kl])\,\Delta^i\Delta^j\Delta^k\Delta^l\right]\\
&\left.+A_i\Delta^i+\frac{\gamma}{2}A_{i,j}\Delta^i\Delta^j-\varepsilon V-\varepsilon(\lambda-\tfrac{1}{12})R\right\}_{q=q_\gamma}.
\end{aligned}
\tag{4.18}
$$

The symbol $\doteq$ denotes equivalence as far as use in the infinitely multiple integral (4.5) is concerned.

At this stage let us compare our method of calculating short time propagators with that found in many papers [7, 8, 10, 14]. There, first the integral representation of the infinitesimal propagator is derived, then the kinetic term is reinstated in the exponential and finally the p integral is performed. In our formulation, that approximation simply amounts to put the operator $\mathscr{L}_\gamma$ in front of the exponential function in (4.6). Obviously, that procedure is less sensitive than our approximation (4.11). The propagators obtained in that way lack the $R_{ij}\Delta^i\Delta^j$ term occuring in (4.17) and S_γ is superseded by

$$
\begin{aligned}
&\tilde{S}_\gamma(q+\Delta,t+\varepsilon;q,t)\\
&=\left\{\frac{1}{2\varepsilon}g_{ij}\Delta^i\Delta^j+(A_i+F_{\gamma i})\Delta^i-\varepsilon(V+(\lambda-\tfrac{1}{12})R+W_\gamma)\right\}_{q=q_\gamma}
\end{aligned}
\tag{4.19}
$$

with additional potentials W_γ and $F_{\gamma i}$ explicitly depending on the parameter γ. They are connected with our expression (4.16) by equivalence relations which hold for the terms in the curly bracket of (4.16)

$$
\begin{aligned}
&\Delta^i\Delta^j\doteq i\varepsilon\, g^{ij}(q_\gamma)\\
&\Delta^i\Delta^j\Delta^k\Delta^l\doteq(i\varepsilon)^2(g^{ij}g^{kl}+g^{ik}g^{jl}+g^{il}g^{jk})_{q=q_\gamma}\\
&\Delta^i\Delta^j\Delta^k\Delta^l\Delta^m\Delta^n\doteq(i\varepsilon)^3(g^{ij}g^{kl}g^{mn}+\text{perm.})_{q=q_\gamma}.
\end{aligned}
\tag{4.20}
$$

(4.20) is obtained from an averaging process in the multiple integral (4.5) with the exponential of (4.16) as the averaging function (cf. [18, 9]). By use of (4.20) W_γ and $F_{\gamma i}$ can be calculated from our formulas. Furthermore, (4.20) shows that Δ may behave like a term of order $\sqrt{\varepsilon}$ in the multiple integral (4.5), as was tacitly minded in the evaluation of (4.15). Although (4.19) may have some virtue in practical calculations, it is not convenient for our purposes. In the next section we show that it is just (4.16) or (4.17) that can be derived in a unique manner from a covariant functional integral.

At the end of this section we call special attention to a nice property of the short time propagator (4.16). Like the exact propagator it satisfies the fixed point property

$$
\begin{aligned}
&K_{s\gamma}(q,t;q',t')\\
&=\int dq''\,g^{1/2}(q'')\cdot K_{s\gamma'}(q,t;q'',t'')\,K_{s\gamma''}(q'',t'';q',t')
\end{aligned}
\tag{4.21}
$$

in all orders given in (4.16). γ, γ' and γ'' are only restricted by (3.3). (4.21) can be verified in a straightforward but tedious calculation. Recently, this property has been invoked by Graham [19] in order to make the short time propagator unique. However, the fixed point property is lost if any term of the bracket in (4.16) is transformed into a potential term via the equivalence relations (4.20).

5. Passage from Lattice Representations to a Covariant Path Integral

In Section 4 the short time propagator (4.17) was directly calculated from the Schrödinger equation (2.5). Let us now approach (4.17) from the opposite side. By extending a treatment of de Witt [15] we show that (4.17) can be obtained in a straightforward way from the classical analog of the quantum

mechanical propagator [20, 15]

$$K_{cl}(q,t;q',t')=G(q,t;q',t')\exp[iS(q,t;q',t')]. \tag{5.1}$$

The phase is the classical action function

$$S(q,t;q',t')=\int_{t'}^{t} dt''\, L(q(t''),\dot{q}(t''))\,dt'' \tag{5.2}$$

with the Lagrangian function

$$L(q,\dot{q})=\tfrac{1}{2}g_{ij}\dot{q}^i\dot{q}^j+A_i\dot{q}^i-V-(\lambda-\tfrac{1}{12})R, \tag{5.3}$$

and the amplitude G, being introduced by Van Vleck [20] in order to have the classical correspondence principle, is given by

$$G(q,t;q',t')$$
$$=(2\pi i)^{-n/2}\,g^{-1/4}(q)\,D^{1/2}(q,t;q',t')\,g^{-1/4}(q'), \tag{5.4}$$

where D is the determinant of the Van Vleck matrix $D_{ij}=-\partial^2 S/\partial q^i\,\partial q'^j$. Further, the classical action principle $\delta S/\delta q=0$ subject to the boundary conditions $q(t')=q'$ and $q(t)=q$ is imposed upon (5.1). Therefore, the action function S may be obtained by solving the equations of motion

$$\ddot{q}^i=-\Gamma^i_{mr}\dot{q}^m\dot{q}^r-g^{ir}(A_{r,m}-A_{m,r})\dot{q}^m$$
$$-g^{ir}(V_{,r}+(\lambda-\tfrac{1}{12})R_{,r}) \tag{5.5}$$

and substituting into the integral (5.2).

Let us now calculate $S(q+\Delta,\ t+\varepsilon;\ q,t)$ as a power series in Δ about $q=q_\gamma$, where q_γ is defined in (4.8). First, by use of the Taylor expansion

$$q^i(t+\tau)-q^i(t)\equiv\vartheta^i$$
$$=\tau\dot{q}^i(t)+\tfrac{1}{2}\tau^2\ddot{q}^i(t)+\tfrac{1}{6}\tau^3\dddot{q}^i(t)+\cdots \tag{5.6}$$

and the equations of motion (5.5) we obtain an expansion of $\dot{q}^i(t)$ in powers of ϑ

$$\tau\dot{q}^i(t)=\{\vartheta^i+\tfrac{1}{2}\Gamma^i_{mn}\vartheta^m\vartheta^n$$
$$+\tfrac{1}{6}(\Gamma^i_{mn,r}+\Gamma^i_{mj}\Gamma^j_{nr})\vartheta^m\vartheta^n\vartheta^r$$
$$+\tfrac{1}{2}g^{ir}(A_{r,n}-A_{n,r})\vartheta^n\tau+\cdots\}_{q=q(t)}. \tag{5.7}$$

Then, this expression is inserted for $q=q_\gamma$, $\tau_\pm=(\gamma\pm1)\,\varepsilon/2$, $\vartheta_\pm=(\gamma\pm1)\,\Delta/2$ into the short time approximation of (5.2)

$$S_\gamma(q+\Delta,t+\varepsilon;q,t)$$
$$=\left\{L(q,\dot{q})(\tau_+-\tau_-)+\frac{1}{2}\frac{dL}{dt}(\tau_+^2-\tau_-^2)+\cdots\right\}_{q=q_\gamma}. \tag{5.8}$$

Here, the second term has to be retained since it contributes to order Δ^2 in S and accordingly survives in the multiple integral (4.5) due to its stochastic nature (see (4.20)). We obtain

$$S_\gamma(q+\Delta,t+\varepsilon;q,t)$$
$$=\left\{\frac{1}{2\varepsilon}g_{ij}[\Delta^i\Delta^j+\gamma\Gamma^j_{mn}\Delta^i\Delta^m\Delta^n\right.$$
$$+\tfrac{1}{12}(1+3\gamma^2)(\Gamma^j_{mn,r}+\Gamma^j_{ms}\Gamma^s_{nr})\Delta^i\Delta^m\Delta^n\Delta^r$$
$$+\tfrac{1}{4}\gamma^2\Gamma^i_{mn}\Gamma^j_{rs}\Delta^m\Delta^n\Delta^r\Delta^s]$$
$$\left.+A_i\Delta^i+\tfrac{1}{2}\gamma A_{i,j}\Delta^i\Delta^j-\varepsilon V-\varepsilon(\lambda-\tfrac{1}{12})R+\cdots\right\}_{q=q_\gamma}, \tag{5.9}$$

which, as can be shown by use of (2.9), is just (4.18). Furthermore, expanding (5.4) in powers of Δ yields [15]

$$G(q+\Delta,t+\varepsilon;q,t)$$
$$=(2\pi i\varepsilon)^{-n/2}\{1+\tfrac{1}{12}R_{ij}\Delta^i\Delta^j+\cdots\}_{q=q_\gamma}. \tag{5.10}$$

Thus we have shown that (4.17) is uniquely determined by (5.1) when the action principle is imposed.

In order to get a path integral functional from (5.1) and (4.5) with the usual measure of integration the amplitude G has to be transformed into an equivalent phase term. There follows from (5.10), (4.20) and (2.8)

$$K(q+\Delta,t+\varepsilon;q,t)\doteq K_F(q+\Delta,t+\varepsilon;q,t)$$
$$=(2\pi i\varepsilon)^{-n/2}\exp[iS_F(q+\Delta,t+\varepsilon;q,t)], \tag{5.11}$$

where covariance is retained and the action S_F is defined in terms of the effective Lagrangian $L_F=L+R/12$.

Now, combining our results (5.11) and (4.5) we finally attain the covariant path integral representation

$$K(q_b,t_b;q_a,t_a)$$
$$=\int_{q(t_a)=q_a}^{q(t_b)=q_b}\mathcal{D}q\,\exp\left[i\int_{t_a}^{t_b}L_F(q(t),\dot{q}(t))\,dt\right]. \tag{5.12}$$

The formal functional integral (5.12) is defined by the time slicing method with the scalar measure of integration

$$\mathcal{D}q=\lim_{N\to\infty}\prod_{j=1}^{N-1}\{dq_j\,g^{1/2}(q_j)(2\pi i\varepsilon)^{-n/2}\}(2\pi i\varepsilon)^{-n/2}. \tag{5.13}$$

Further, the requirement is imposed that in every time intervall the action is calculated for a true classical trajectory. By this the quantum theory is uniquely defined without any reference to a specifically chosen lattice representation.

At the end of this section, let us discuss the curvature potential λR in (2.7). Various factors λ are floating around in the literature [7, 8, 13, 21]. Obviously, R should not occur in the classical Lagrangian, since it

is proportional to $\hbar^2$. If Feynman's quantization method is taken as a postulate, then L_F is supposed to be independent of R and consequently $\lambda=\lambda_F\equiv 1/6$ [21]. However, some arguments are given now which may indicate to favour L as the classical Lagrangian and accordingly $\lambda=\lambda_V\equiv 1/12$.

1. The term $R_{ij}\Delta^i\Delta^j$ in (4.17) is completely fixed by the action (5.2), as follows from (5.4) and (5.10). It has a natural origin in a transformation from variables q, p to $q, q+\Delta$ [20, 15]. Therefore, dealing with this term seperately is more reasonable than merging it in the definition of the action.

2. The fixed point property (4.21) doesn't hold for K_F.

3. If $n=4$, (2.5) may be a representation of the Klein-Gordon or Dirac equation with an electromagnetic interaction in Fock's fifth-parameter form. In this case, t is a redundant parameter being singled out at the end. Relying on L as the classical Lagrangian ($\lambda=\lambda_V$) with $V=\mu_0^2/2$ leads to the covariant Klein-Gordon equation

$$[g^{-1/4}(\mathbf{p}_i-A_i)\,g^{ij}\,g^{1/2}(\mathbf{p}_j-A_j)\,g^{-1/4}+\mu_0^2+R/6]\,\Psi=0 \tag{5.14}$$

with two nice properties due to the term $R/6$. First, $R/6$ is just the term necessary for conformal invariance of (5.14) in the massless case [19, 22]. The second virtue of (5.14) has been enlightened by Graham [12]. If (5.14) is coupled to the gravitational field, a well behaved energy-momentum tensor as a source of gravity is obtained. This tensor, originally introduced by Callan et al. [23] in an ad hoc manner in order to get rid of some nongraviational divergencies, appears in a natural way when L is supposed to be the underlying classical Lagrangian.

6. Application to Diffusion Processes

Using the formal analogy between the Schrödinger equation and the Fokker-Planck equation we can immediately apply the results of the last section to the general nonlinear diffusion process. In our notation of Sect. 2 the covariant Fokker-Planck equation [24] reads

$$\frac{\partial}{\partial\tau}P(x,\tau)=\{-\tfrac{1}{2}Q^{1/4}\,\mathbf{p}_i\,Q^{ij}\,Q^{-1/2}\,\mathbf{p}_j\,Q^{1/4} - iQ^{1/4}\,\mathbf{p}_i\,h^i\,Q^{-1/4}\}\,P(x,\tau), \tag{6.1}$$

where

$$\mathbf{p}_i=-iQ^{1/4}(x)\frac{\partial}{\partial x^i}Q^{-1/4}(x). \tag{6.2}$$

Comparing (6.1) with (2.5) and (2.7) we may establish the following dictionary for translating the quantum mechanical into statistical quantities

$$\begin{aligned}
&t\to i\tau && \tau=\text{time}\\
&q^i\to x^i && x^i=\text{complete set of macrovariables}\\
&g^{ij}\to -Q^{ij} && Q^{ij}=\text{diffusion matrix, } Q=|Q^{ij}|\\
&A_i\to -iQ_{ij}\,h^j && h_i=\text{covariant driftvector}\\
&V\to -\tfrac{1}{2}h^i\,Q_{ij}\,h^j-\tfrac{1}{2}Q^{1/2}(h^i\,Q^{-1/2})_{,\,i}+(\lambda-1/12)\,R\\
&\Psi\to P && P=\text{probability density.}
\end{aligned} \tag{6.3}$$

The probability density is a scalar of weight zero normalized according to

$$\int dx\,Q^{-1/2}(x)\,P(x,\tau)=1. \tag{6.4}$$

The fundamental solution of (6.1) is the invariant conditional probability density $P(x,\tau;x',\tau')$ with the property

$$P(x,\tau;x',\tau)=Q^{1/2}(x)\,\delta(x-x'). \tag{6.5}$$

There follows from (5.12) and (6.3) the path integral representation

$$P(x,\tau;x',\tau') = \int_{x(\tau')=x'}^{x(\tau)=x}\mathscr{D}x\,\exp\left[-\int_{\tau'}^{\tau}d\tau''\,L_F(x(\tau''),\dot{x}(\tau''))\right], \tag{6.6}$$

with the scalar measure of integration ($\varepsilon=(\tau-\tau')/N$)

$$\mathscr{D}x=\lim_{N\to\infty}\prod_{j=1}^{N-1}\{dx_j/\sqrt{(2\pi\varepsilon)^n\,Q(x_j)}\}\,(2\pi\varepsilon)^{-n/2} \tag{6.7}$$

and $L_F=L+\dfrac{R}{12}$, where L is the Lagrangian ("Onsager Machlup function") for diffusion

$$L(x,\dot{x})=\tfrac{1}{2}Q_{ij}(\dot{x}^i-h^i)(\dot{x}^j-h^j) +\tfrac{1}{2}Q^{1/2}(h^i\,Q^{-1/2})_{,\,i}+R/12. \tag{6.8}$$

R is the curvature scalar in phase space defined by (2.8) and (6.3).

Like (5.12) the formal functional integral (6.6) is defined by the time slicing method and the requirement that in each time intervall the action principle holds. By this the nonlinear diffusion process is uniquely defined without reference to any specific lattice choice.

7. Conclusions

We have studied the interrelation of equations of motion and path integrals in coordinate space both for nonlinear quantum mechanics and diffusion processes. The connection between operator ordering schemes

and lattice representations of path integrals was investigated and a covariant Lagrangian was extracted. Our Lagrangian for diffusion agrees with Graham's [13]. In Graham's paper the Lagrangian is obtained from the short time propagator by the continuum transition $\Delta^i \to \varepsilon \dot{q}^i$. However, the discretization process in the reverse direction is not unique and has to be supplemented by the fixed point property of the short time propagator [19]. In our treatise the uniqueness in both directions is ensured by the action principle.

A more general class of equivalent lattice representations is obtained by forming normalized linear combinations of the short time action S_γ or $\tilde{S}_\gamma$ as defined in (4.18) and (4.19) with different γ, e.g.

$$S=\sum_i a_i S_{\gamma_i}; \quad \tilde{S}=\sum_i a_i \tilde{S}_{\gamma_i}; \quad \sum_i a_i=1, \quad a_i \geqq 0. \quad (7.1)$$

In (7.1) that class which was recently discussed by Haken [25] is contained. Of course, (7.1) is connected with a more general class of operator ordering prescriptions than discussed in this paper.

Surveying our treatise a foolproof route in performing nonlinear point transformations $q'^i=f^i(q)$ is advocated. Clearly, the transformation is most easily performed in the covariant formal functional integral (5.12) or (6.6). Then, any specific lattice representation and the noncovariant additional potential terms W_γ and $F_{\gamma i}$ may be calculated according to our rules of the preceeding sections.

As to perturbation theory, the Feynman rules will explicitly depend on the discretization (operator ordering) parameters. However, in any particular scheme a more or less intricate cancellation mechanism between individual diagrams will take place which ensures that the complete theory is independent of the particular discretization chosen. Recently, this has been confirmed in explicit calculations for a reducible system [26].

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